

SIMATS ENGINEERING

Department of Computer Science and Engineering



CSA15 Cloud Computing and Big Data Analytics CO2 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in CO1 and CO2 AT1
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from CO1 and CO2 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table. • Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM

Preferred cloud evidence:
subscription, VM, SSH, NGINX/test
service, firewall, shutdown

Track B - Local VM

VirtualBox or VMware with Tiny
Core, Alpine, Debian netinst or
Ubuntu Server

Track C - Docker / Podman

Run a lightweight service, map a
port, inspect logs, compare with
VM

Track D - Simulation

Used only if deployment is
blocked; must include technical
configuration and security
evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository
link and GCR submission.

Technical Orientation

The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why
it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	Karanam Reddy Vishnu Chaithanya
Register Number	193272057
Class / Slot	Slot B

Capstone Title	Sentinal X
CO2 AT1 Recommended VM Size	vCPU: ____ RAM: ____ Disk: ____
Selected Practical Track	Azure VM / Local VM / Docker / Simulation
GitHub Repository Link	
Google Classroom Submission Date	

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Workload type	
Main application modules	
Expected users / traffic assumption	
Database need	
File storage need	
AI/Python/analytics need	
Security concern	
Recovery concern	

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

	Approximate Download Size /		
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command line practice and container-like thinking.	https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/
Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download

Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server
Selection Guidance For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.			
I have selected an OS image suitable for my device and network.	☐ Yes ☐ Rework		
I have recorded the official download page or cloud image name.	☐ Yes ☐ Rework		
I have checked whether the ISO is small enough for the lab duration.	☐ Yes ☐ Rework		

Student Entry: Selected OS image, version, reason and link used

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95      88.0K     88.0K     0 100% /tap/tcloop/iw
/dev/loop96      30.7M     30.7M     0 100% /tap/tcloop/firmware-atheros
/dev/loop97      275.0K    275.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98      292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M     0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M     0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K     0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K     32.0K     0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K     92.0K     0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.2M      3.2M      0 100% /tap/tcloop/hwdata
/dev/loop114      36.0K     36.0K     0 100% /tap/tcloop/libpci
/dev/loop115      76.0K     76.0K     0 100% /tap/tcloop/pclutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K     0 100% /tap/tcloop/firmware-rtl8139
/dev/loop119     64.0K     64.0K     0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K     0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K     0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K     0 100% /tap/tcloop/advcpmp
/dev/loop123     452.0K    452.0K     0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K     0 100% /tap/tcloop/e2fsprogs
/dev/loop125     20.8M     20.8M     0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

SIMATS Engineering | Department of Computer Science and Engineering | Official Assessment Material

CSA15 | Cloud Computing and Big Data Analytics CO2 AT2 | Student Workbook

Part 3: Track A - Azure Linux VM Guided Procedure

Preferred Cloud Path

Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

A. Account Activation and Portal Access

1. Open the Azure for Students page and choose the student sign-up option.

2. Sign in using the official student email account when available.
3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
4. Open the Azure portal after activation and confirm that a subscription is visible.
5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account information.

Azure portal opens successfully.	☐ Yes ☐ Rework	
A student or valid Azure subscription is visible.	☐ Yes ☐ Rework	
The student understands that resources must be stopped or deleted after evidence capture.	☐ Yes ☐ Rework	
Evidence Space: Paste account activation/portal screenshot with sensitive details hidden		

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95     88.0K     88.0K    0 100% /tap/tcloop/iw
/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K    276.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M    0 100% /tap/tcloop/firmware-iwlwifi
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/dev/loop106     16.0K     16.0K    0 100% /tap/tcloop/firmware-openwif
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/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-uxge
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/dev/loop113      3.3M      3.3M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K    0 100% /tap/tcloop/pciutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-ttl_nic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K    0 100% /tap/tcloop/advcpmp
/dev/loop123     452.0K    452.0K    0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
/dev/loop125     20.8M     20.8M    0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:37:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

B. Create the Linux Virtual Machine

6. In Azure portal, search for Virtual machines and choose Create.
7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno. 8. Enter a VM name such as csa15-vm-yourregno.
9. Select a nearby Azure region available in the subscription.
10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image. 11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.

12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.
14. Review and create the VM. Wait until deployment succeeds.

Resource group		
VM name		
Region		
Image		
VM size		
vCPU		
RAM		
Disk type and size		
Authentication method		SSH key / password - do not paste secret
Open inbound ports		
VM deployment status shows success.	☐ Yes ☐ Rework	
VM overview page shows running status.	☐ Yes ☐ Rework	
Public IP or connection method is visible.	☐ Yes ☐ Rework	
The selected VM size is compared with CO2 AT1 sizing recommendation.	☐ Yes ☐ Rework	
Evidence Space: Paste VM overview and configuration screenshot		

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
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/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
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/dev/loop98     292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M    0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M    0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K    0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K     32.0K    0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K     92.0K    0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.3M      3.3M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K    0 100% /tap/tcloop/pclutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-ttl_nic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K    0 100% /tap/tcloop/advcpmp
/dev/loop123     452.0K    452.0K    0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
/dev/loop125     20.8M     20.8M    0 100% /tap/tcloop/firmware-andgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G          108.1M          1.8G          27.8M          117.8M          1.7G
Swap:          486.6M           0          486.6M

```

C. Connect and Verify the VM

15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
17. After login, run: `uname -a`
18. Run: `lsb_release -a` or `cat /etc/os-release`
19. Run: `free -h`
20. Run: `df -h`
21. Run: `ip addr` or `hostname -I`
22. Record the output as screenshot evidence.

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uname -a	Kernel and OS environment	
cat /etc/os-release	Linux distribution details	
free -h	RAM allocation	
df -h	Disk allocation	
hostname -I	Private IP / network evidence	

Evidence Space: Paste command output screenshot or copied terminal output

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     364.0K    364.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
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/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
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/dev/loop113     3.2M      3.2M      0 100% /tap/tcloop/hwdata
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/dev/loop117     4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-mtlncic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
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/dev/loop124     24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
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tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

D. Optional Web Service Check

23. Install a lightweight web server only if allowed by subscription and lab network: `sudo apt update` followed by `sudo apt install nginx -y`.
24. Start or verify the service: `systemctl status nginx`.
25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.
26. If port 80 is not opened, record the reason and keep SSH-only evidence.

The student can explain why a public port was opened or not opened.	☐ Yes ☐ Rework	
The service status or browser test was captured.	☐ Yes ☐ Rework	
The public exposure was kept temporary for the assessment.	☐ Yes ☐ Rework	

E. Azure Security and Cost-Control Closeout

27. Review Networking and record inbound port rules.
28. Confirm that no unnecessary public ports are open.
29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
30. Capture stopped/deallocated or deletion evidence.
31. Record what would change for a production capstone deployment.

Inbound ports allowed	
Authentication method used	
Public IP required?	
Stopped/deallocated or deleted?	
Reason for closeout choice	
Evidence Space: Paste networking rule and shutdown/deletion screenshot	

```
Terminal
/dev/loop91      2.9M    2.9M    0 100% /tap/tcloop/openssl
/dev/loop92     354.0K  354.0K  0 100% /tap/tcloop/libnl
/dev/loop93      1.3M    1.3M    0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K    8.0K    0 100% /tap/tcloop/wifi
/dev/loop95      88.0K   88.0K   0 100% /tap/tcloop/tw
/dev/loop96     30.7M   30.7M   0 100% /tap/tcloop/firmware-atheros
/dev/loop97     275.0K  275.0K  0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K  292.0K  0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M    2.4M    0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M    3.9M    0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M    5.6M    0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M   38.8M   0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M   80.6M   0 100% /tap/tcloop/firmware-avril
/dev/loop104      1.9M    1.9M    0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M    1.4M    0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K   16.0K   0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K   32.0K   0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K 1000.0K  0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M    5.1M    0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M    1.1M    0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K  876.0K   0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K   92.0K   0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.2M    3.2M    0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K   36.0K   0 100% /tap/tcloop/libpci
/dev/loop115     76.0K   76.0K   0 100% /tap/tcloop/pciutils
/dev/loop116      3.3M    3.3M    0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K    4.0K    0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K   56.0K   0 100% /tap/tcloop/firmware-rtl8139
/dev/loop119     64.0K   64.0K   0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K   56.0K   0 100% /tap/tcloop/gzip
/dev/loop121     196.0K  196.0K   0 100% /tap/tcloop/tar
/dev/loop122     228.0K  228.0K   0 100% /tap/tcloop/advcpmp
/dev/loop123     452.0K  452.0K   0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K   24.0K   0 100% /tap/tcloop/ezra-master
/dev/loop125     20.8M   20.8M   0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0:  Link encap:Ethernet  HWaddr 08:00:27:7C:37:A3
        inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:11 errors:0 dropped:0 overruns:0 frame:0
        TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo:     Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        UP LOOPBACK RUNNING  MTU:65536  Metric:1
        RX packets:4 errors:0 dropped:0 overruns:0 frame:0
        TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M         1.7G
Swap:          486.5M           0         486.5M
```

Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Hypervisor selected	VirtualBox / VMware Workstation / VMware Fusion	
Version		
Host operating system	Windows / macOS / Linux	
Host CPU and RAM		
Reason for selected hypervisor		
Hypervisor opens successfully.	☐ Yes ☐ Rework	
Student can locate New VM/Create VM option.	☐ Yes ☐ Rework	
Student has downloaded or selected a suitable OS image.	☐ Yes ☐ Rework	

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on CO2 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on CO2 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

VM name	csa15-localvm-regno	
OS image	Tiny Core / Alpine / Debian / Ubuntu	
vCPU	1-2 for lightweight OS; 2+ if host allows	
RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	

Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	
Network mode	NAT for normal lab use	

Display / boot mode	Default unless OS requires change	
---------------------	-----------------------------------	--

Host Safety Guidance

Do not allocate all host resources to the VM. Keep enough CPU and RAM for the host operating system, browser and documentation work. If the host has 8 GB RAM, a 1 GB or 2 GB VM is more appropriate than a 6 GB VM.

Evidence Space: Paste VM settings screenshot before first boot

```
Terminal
/dev/loop91      2.9M    2.9M    0 100% /tap/tcloop/openssl
/dev/loop92     364.0K  364.0K  0 100% /tap/tcloop/libnl
/dev/loop93      1.3M    1.3M    0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K    8.0K    0 100% /tap/tcloop/wifi
/dev/loop95     88.0K   88.0K    0 100% /tap/tcloop/iw
/dev/loop96     30.7M   30.7M    0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K  276.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K  292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M    2.4M    0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M    3.9M    0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M    5.6M    0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M   38.8M    0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M   80.6M    0 100% /tap/tcloop/firmware-marvell
/dev/loop104      1.9M    1.9M    0 100% /tap/tcloop/firmware-mwl10ge
/dev/loop105      1.4M    1.4M    0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K   16.0K    0 100% /tap/tcloop/firmware-openwulf
/dev/loop107     32.0K   32.0K    0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K 1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M    5.1M    0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M    1.1M    0 100% /tap/tcloop/firmware-usb-ata
/dev/loop111     876.0K  876.0K    0 100% /tap/tcloop/firmware-vxge
/dev/loop112     92.0K   92.0K    0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.2M    3.2M    0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K   36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K   76.0K    0 100% /tap/tcloop/pciutils
/dev/loop116      3.3M    3.3M    0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K    4.0K    0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K   56.0K    0 100% /tap/tcloop/firmware-rtl_nic
/dev/loop119     64.0K   64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K   56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K  196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K  228.0K    0 100% /tap/tcloop/advcomp
/dev/loop123     452.0K  452.0K    0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K   24.0K    0 100% /tap/tcloop/ezra-master
/dev/loop125     20.8M   20.8M    0 100% /tap/tcloop/firmware-sedgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.0G         106.1M         1.8G         27.8M        117.8M         1.7G
Swap:          486.6M           0         486.6M
```

C. Boot and Verify the VM

- 45. Start the VM and wait for the OS boot screen.
- 46. For Tiny Core, confirm that the graphical desktop appears.
- 47. For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
- 48. For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
- 49. Open terminal inside the VM where possible.
- 50. Run or record equivalent commands: uname -a, free -h, df -h and ip addr.

VM starts without boot error.	☐ Yes ☐ Rework	
OS screen or terminal is visible.	☐ Yes ☐ Rework	
CPU/RAM/disk allocation is verified.	☐ Yes ☐ Rework	
Network mode is recorded.	☐ Yes ☐ Rework	
Evidence Space: Paste VM boot/login/terminal screenshots		

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95     88.0K     88.0K    0 100% /tap/tcloop/iw
/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K    276.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M    0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M    0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K    0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K     32.0K    0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K     92.0K    0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.3M      3.3M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K    0 100% /tap/tcloop/pclutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-tl1nic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K    0 100% /tap/tcloop/advccp
/dev/loop123     452.0K    452.0K    0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
/dev/loop125     20.8M     20.8M    0 100% /tap/tcloop/firmware-andgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

D. Snapshot and Clone Practice

51. Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
52. Create a snapshot named before-update-or-demo.
53. Record snapshot name, date and reason.
54. Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
55. Start the VM again and confirm it still boots.

Snapshot name	
Snapshot timing	

Snapshot reason	
Clone attempted?	Yes / No
Clone type	Full / Linked / Not attempted
Storage impact observed	

Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot

```

Terminal
/dev/loop91      2.9M    2.9M    0 100% /tap/tcloop/openssl
/dev/loop92     364.0K  364.0K  0 100% /tap/tcloop/libnl
/dev/loop93      1.3M    1.3M    0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K    8.0K    0 100% /tap/tcloop/wifi
/dev/loop95      88.0K   88.0K   0 100% /tap/tcloop/tw
/dev/loop96      30.7M   30.7M   0 100% /tap/tcloop/firmware-atheros
/dev/loop97      276.0K  276.0K  0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98      292.0K  292.0K  0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M    2.4M    0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M    3.9M    0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M    5.6M    0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M   38.8M   0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M   80.6M   0 100% /tap/tcloop/firmware-marvel
/dev/loop104     1.5M    1.5M    0 100% /tap/tcloop/firmware-myr110ge
/dev/loop105     1.4M    1.4M    0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K   16.0K   0 100% /tap/tcloop/firmware-openfwaf
/dev/loop107     32.0K   32.0K   0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K 1000.0K  0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109     5.1M    5.1M    0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110     1.1M    1.1M    0 100% /tap/tcloop/firmware-uagle-ata
/dev/loop111     876.0K  876.0K   0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K   92.0K   0 100% /tap/tcloop/firmware-zd1211
/dev/loop113     3.2M    3.2M    0 100% /tap/tcloop/hwdmdata
/dev/loop114     36.0K   36.0K   0 100% /tap/tcloop/libpci
/dev/loop115     76.0K   76.0K   0 100% /tap/tcloop/pciutils
/dev/loop116     3.3M    3.3M    0 100% /tap/tcloop/firmware-chelsio
/dev/loop117     4.0K    4.0K    0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K   56.0K   0 100% /tap/tcloop/firmware-rtl_nic
/dev/loop119     64.0K   64.0K   0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K   56.0K   0 100% /tap/tcloop/gzip
/dev/loop121     196.0K  196.0K   0 100% /tap/tcloop/tar
/dev/loop122     228.0K  228.0K   0 100% /tap/tcloop/advcomp
/dev/loop123     452.0K  452.0K   0 100% /tap/tcloop/mkisoofs-tools
/dev/loop124     24.0K   24.0K   0 100% /tap/tcloop/ezraester
/dev/loop125     20.8M   20.8M   0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:37:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:111 errors:0 dropped:0 overruns:0 frame:0
          TX packets:111 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M         1.7G
Swap:          486.6M           0         486.6M

```

E. Local VM Security Reflection

VM network mode	NAT / Bridged / Host-only / Other
Why this mode is selected	
Guest password policy	
Shared folders enabled?	Yes / No
Clipboard/drag-drop enabled?	Yes / No
Risk if VM is exposed to public network	

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: `docker --version`.
58. Run a lightweight container such as: `docker run --rm hello-world`.
59. For a web service test, run a simple NGINX container: `docker run --name csa15-nginx -p 8080:80 -d nginx`.
60. Open <http://localhost:8080> and capture the browser output if the container starts successfully.
61. Check logs using: `docker logs csa15-nginx`.
62. Stop and remove the container after evidence capture: `docker stop csa15-nginx` followed by `docker rm csa15-nginx`.

docker --version output is captured.	☐ Yes ☐ Rework	
A container run command is captured.	☐ Yes ☐ Rework	
Port mapping or service evidence is captured if web test	☐ Yes ☐ Rework	

is used.		
Container logs or status are captured.	☐ Yes ☐ Rework	
Container is stopped/removed after evidence capture.	☐ Yes ☐ Rework	
Docker version		
Image used		
Container name		
Port mapping		
Service URL tested		
Log/status observation		
Stop/remove evidence		

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     364.0K    364.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95     88.0K     88.0K      0 100% /tap/tcloop/iw
/dev/loop96     30.7M     30.7M      0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K    276.0K      0 100% /tap/tcloop/firmware-ips2100
/dev/loop98     292.0K    292.0K      0 100% /tap/tcloop/firmware-ips2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M      0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M      0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K      0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K     32.0K      0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K      0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K      0 100% /tap/tcloop/firmware-vxge
/dev/loop112     92.0K     92.0K      0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.3M      3.3M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K      0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K      0 100% /tap/tcloop/pclutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K      0 100% /tap/tcloop/firmware-rtl8139
/dev/loop119     64.0K     64.0K      0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K      0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K      0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K      0 100% /tap/tcloop/advcpmp
/dev/loop123     452.0K    452.0K      0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K      0 100% /tap/tcloop/ezraester
/dev/loop125     20.8M     20.8M      0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:37:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G          108.1M          1.8G          27.8M          117.8M          1.7G
Swap:          486.6M           0          486.6M

```

B. Capstone Module Decision

Frontend		

Backend/API		
Database		
File storage		
AI/Python module		
Analytics/dashboard		
Notification/background job		

C. Optional Container and Orchestration Extension

Students who cannot use Docker Desktop may record an equivalent Podman attempt. Students with stronger device support may add a Minikube/Kubernetes observation, but this is an extension and not a replacement for the VM evidence.

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes documentation: <https://kubernetes.io/docs/home/>

Podman	podman --version, image pull/run, logs, stop/remove evidence	
Minikube	minikube start/status, kubectl get nodes/pods evidence	
Kubernetes	Simple deployment/service YAML or command evidence if attempted	

Part 6: Track D - Technical Simulation Path

Use Only When Required

This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

CloudSim: <https://github.com/Cloudslab/cloudsim>

Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.
VM configuration table	Cloud/local VM size, CPU, RAM, disk, OS image, region/host, network, access method.
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.
Snapshot/clone plan	Names, timing, retention and rollback purpose.
Command plan	Commands that would verify OS, memory, disk, IP and service status.

Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.
Blocker Evidence and Justification	
Platform	Azure / VirtualBox / VMware / Docker
OS image	
vCPU	
RAM	
Disk	
Network mode / firewall	
Access method	
Service to verify	
Paste or draw the simulation architecture diagram here	

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95     88.0K     88.0K    0 100% /tap/tcloop/iw
/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K    276.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M    0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M    0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-mgri10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K    0 100% /tap/tcloop/firmware-openfwaf
/dev/loop107     32.0K     32.0K    0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109     5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-usagie-ata
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-vcge
/dev/loop112     92.0K     92.0K    0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.2M      3.2M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K    0 100% /tap/tcloop/pciutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-rtl_nic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K    0 100% /tap/tcloop/advccp
/dev/loop123     452.0K    452.0K    0 100% /tap/tcloop/mkisoofs-tools
/dev/loop124      24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
/dev/loop125     20.8M     20.8M    0 100% /tap/tcloop/firmware-amdgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:B7:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

Part 7: CO2 Reflection and Infrastructure Defense

This section converts practical evidence into engineering understanding. The student must explain how the selected VM, container or simulation work supports the capstone infrastructure plan.

What CO2 concept became clearer through this activity?	
--	--

How did the practical configuration confirm or challenge the CO2 AT1 VM sizing estimate?	
Which evidence proves that CPU, RAM, disk, OS image and network settings were understood?	
Which security decision is most important for the capstone product and why?	
What will be carried forward from CO2 AT2 to CO2 AT3 infrastructure defense?	
Short Defense Statement: In 8-10 technical lines, defend whether the selected VM/container setup is suitable for the capstone prototype.	

```

Terminal
/dev/loop91      2.9M      2.9M      0 100% /tap/tcloop/openssl
/dev/loop92     354.0K    354.0K    0 100% /tap/tcloop/libnl
/dev/loop93      1.3M      1.3M      0 100% /tap/tcloop/wpa_supplicant-dbus
/dev/loop94      8.0K      8.0K      0 100% /tap/tcloop/wifi
/dev/loop95     88.0K     88.0K    0 100% /tap/tcloop/iw
/dev/loop96     30.7M     30.7M    0 100% /tap/tcloop/firmware-atheros
/dev/loop97     276.0K    276.0K    0 100% /tap/tcloop/firmware-ipw2100
/dev/loop98     292.0K    292.0K    0 100% /tap/tcloop/firmware-ipw2200
/dev/loop99      2.4M      2.4M      0 100% /tap/tcloop/firmware-iwlmax
/dev/loop100     3.9M      3.9M      0 100% /tap/tcloop/firmware-iwl8000
/dev/loop101     5.6M      5.6M      0 100% /tap/tcloop/firmware-iwl9000
/dev/loop102     38.8M     38.8M    0 100% /tap/tcloop/firmware-iwlwifi
/dev/loop103     80.6M     80.6M    0 100% /tap/tcloop/firmware-marvel
/dev/loop104      1.9M      1.9M      0 100% /tap/tcloop/firmware-myl10ge
/dev/loop105      1.4M      1.4M      0 100% /tap/tcloop/firmware-netxen
/dev/loop106     16.0K     16.0K    0 100% /tap/tcloop/firmware-openwif
/dev/loop107     32.0K     32.0K    0 100% /tap/tcloop/firmware-ralinkwifi
/dev/loop108    1000.0K   1000.0K    0 100% /tap/tcloop/firmware-rtlwifi
/dev/loop109      5.1M      5.1M      0 100% /tap/tcloop/firmware-ti-connectivity
/dev/loop110      1.1M      1.1M      0 100% /tap/tcloop/firmware-ueagle-atm
/dev/loop111     876.0K    876.0K    0 100% /tap/tcloop/firmware-uxge
/dev/loop112     92.0K     92.0K    0 100% /tap/tcloop/firmware-zd1211
/dev/loop113      3.3M      3.3M      0 100% /tap/tcloop/hwdata
/dev/loop114     36.0K     36.0K    0 100% /tap/tcloop/libpci
/dev/loop115     76.0K     76.0K    0 100% /tap/tcloop/pclutils
/dev/loop116      3.3M      3.3M      0 100% /tap/tcloop/firmware-chelsio
/dev/loop117      4.0K      4.0K      0 100% /tap/tcloop/firmware-intel_e100
/dev/loop118     56.0K     56.0K    0 100% /tap/tcloop/firmware-tl1nic
/dev/loop119     64.0K     64.0K    0 100% /tap/tcloop/firmware-tigon
/dev/loop120     56.0K     56.0K    0 100% /tap/tcloop/gzip
/dev/loop121     196.0K    196.0K    0 100% /tap/tcloop/tar
/dev/loop122     228.0K    228.0K    0 100% /tap/tcloop/advccp
/dev/loop123     452.0K    452.0K    0 100% /tap/tcloop/wkisoofs-tools
/dev/loop124      24.0K     24.0K    0 100% /tap/tcloop/ezrenaster
/dev/loop125     20.8M     20.8M    0 100% /tap/tcloop/firmware-andgpu

tc@box:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7C:37:A3
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:11 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2003 (1.9 KiB)  TX bytes:1418 (1.3 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:200 (200.0 B)  TX bytes:200 (200.0 B)

tc@box:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           2.0G         108.1M         1.8G         27.8M        117.8M        1.7G
Swap:          486.6M           0         486.6M

```

Student-Facing Assessment Criteria

Capstone continuity	Evidence connects clearly to the same capstone title and CO2 AT1 sizing decision.
Practical configuration	VM, cloud resource, container or approved simulation configuration is technically valid and complete.
Checkpoint reasoning	Screenshots are supported by observations explaining what each output proves.
Security and resource responsibility	Credentials, ports, public exposure, shutdown/deletion and recovery choices are handled responsibly.
Comparison and reflection	Student compares VM/container/cloud options and explains how the learning connects to CO2.
Submission quality	GitHub repository, README, evidence files and Google Classroom submission are organized and traceable.

Part 8: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

README.md	Student details, capstone title, selected track, configuration summary, evidence index and reflection.	
Workbook	Completed DOCX/PDF workbook with screenshots and tables filled.	
Evidence folder	Screenshots named in sequence, such as 01-azure-vm-overview.png.	
Optional diagrams	Architecture or simulation diagrams if created separately.	
Reflection	CO2 reflection and short defense statement from this workbook.	
1		
2		
3		
4		
5		
6		
7		
8		
Repository is created under student's GitHub account.	☐ Yes ☐ Rework	
README.md is complete.	☐ Yes ☐ Rework	
Workbook is uploaded.	☐ Yes ☐ Rework	
Screenshots are uploaded and named clearly.	☐ Yes ☐ Rework	
Google Classroom submission contains the GitHub repository link.	☐ Yes ☐ Rework	
No passwords, private keys, tokens or sensitive account details are uploaded.	☐ Yes ☐ Rework	

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Karanam Reddy Vishnu Chaithanya
Register Number	192372057
Signature	
Date	

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	

Reference Links Used in This Workbook

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

Tiny Core Linux downloads: <https://distro.ibiblio.org/tinycorelinux/downloads.html>

Alpine Linux downloads: <https://alpinelinux.org/downloads/>

Alpine virt ISO directory: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst: <https://www.debian.org/download>

Debian network install: <https://www.debian.org/CD/netinst/>

Ubuntu Server: <https://ubuntu.com/download/server>

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes: <https://kubernetes.io/docs/home/>

CloudSim: <https://github.com/Cloudslab/cloudsim>

